



**DATA ANALYSIS AND VISUALIZATION
SESSIONAL TERM ASSESSMENT**

CASE STUDY

**Case Study Proposal: U.S. Crime Data Analysis (State-Level,
2000–2022)**

Group Members:

Noor-ul-Huda 23i-2099

Malaika Zainab 23i-2605

Maimoona Hanan 23i-2544

Submitted to: Ma'am Maryam Hussain

Name of Dataset: National Incident Based Reporting System (NIBRS) Database (2000 – 2022)

Dataset 2: State By Date Level

Dataset Link: https://dasil.sites.grinnell.edu/downloadable-data/?utm_source

National Incident Based Reporting System (NIBRS) Database (2000 – 2022)

These data from NIBRS include the nature and types of specific offenses in the incident, characteristics of victim(s) and offender(s), and types and values of property stolen and recovered.

Dataset 1: Individual Incident Level

Download: [CSV](#) (2.1GB) [STATA](#) (2.3GB) [Codebook](#)

Dataset 2: State By Date Level

Download: [CSV](#) (5.0MB) [STATA](#) (20.5MB) [Codebook](#)

Drive link:

https://docs.google.com/document/d/1HnHxCC8zmK8cIm8_0SWKaawKgpKCN7rELPA9KUMBcUg/edit?usp=drive_link

1. Background and Context

Crime is a critical social issue that affects communities, policy, and public safety. Understanding crime patterns over time and across geographic regions can help law enforcement, policymakers, and social researchers implement effective interventions and preventive measures.

The National Incident-Based Reporting System (NIBRS), maintained by the FBI, collects detailed crime data from law enforcement agencies across the United States. Unlike aggregated crime summaries, NIBRS captures individual incident-level details, including types of offenses, victim and offender demographics, and property-related losses.

This case study uses Dataset 2: State-By-Date Level, which summarizes incidents for each U.S. state on each date. The dataset spans 2000–2022 and contains totals for offenses, victims, offenders, and subcategories such as violence, theft, drug-related crimes, sex offenses, gun involvement, and property losses.

2. Case Study Problem

Although raw crime data has existed for decades, understanding temporal and geographic patterns is complex. Law enforcement agencies and policymakers often lack insights into how crime trends vary across states and over time or how factors like gun involvement, drug activity, and offender demographics influence crime.

4. Problem Statement:

“How have crime patterns evolved across U.S. states from 2000 to 2022, and what demographic, geographic, and offense-type factors influence these trends?”

5. Research Questions

1. How do total crimes, victims, and offenders vary by state and year?
2. Which types of crimes (violent, theft, drug, sex offenses) are most prevalent in specific states?
3. What trends exist in gun-involved or drug-related crimes over time?
4. How do offender and victim demographics (gender, race, age) vary across states and crime types?
5. Are there correlations between offense types, property loss, and demographic factors?

6. Dataset Description

Feature	Description
state	Two-letter U.S. state abbreviation (e.g., AL, CA, TX)
date_HRF / year	Date in YYYYMMDD format and the corresponding year
total_offense / total_victim / total_offender	Aggregate daily counts of offenses, victims, and offenders
violence_offense / theft_offense / drug_offense / sex_offense / kidnapping_trafficking / other_offense	Type-specific crime counts
gun_involvement, completed, attempted	Crime characteristics
property_value, stolen_motor	Monetary value of property loss and motor vehicle theft counts
male_offender, female_offender, w_offender, b_offender, i_offender, a_offender, p_offender, unknown_race_offender	Offender demographic breakdowns
male_victim, female_victim, w_victim, b_victim, i_victim, a_victim, p_victim, unknown_race_victim	Victim demographic breakdowns
offender_wi_family / offender_outside_family / offender_not_known	Relationship context between offender and victim
minor_victim / non_minor_victim	Victim age categories

The dataset provides a structured and summarized view of crime incidents at the state and date level, enabling both temporal (time-series) and geographic (state-level) analyses.

7. Objectives

The study aims to:

1. Analyze state-level trends in total offenses, victims, and offenders.
2. Explore demographic patterns among offenders and victims (gender, age, race).
3. Examine correlations between offense types, gun involvement, and property losses.
4. Identify states and years with high or low crime prevalence.
5. Provide insights that can inform social policy, law enforcement strategies, and community awareness.

8. Tools and Environment

All analyses will be conducted using **Python libraries**

9. Methodology

Step 1: Data Preparation

- Load CSV data into Python using pandas.
- Handle missing values, zeros, or inconsistent data entries.
- Convert date columns to datetime objects.

- Create derived variables (monthly/annual totals, offense ratios, etc.).

Step 2: Exploratory Data Analysis (EDA)

- Calculate summary statistics for offense types, demographics, and property losses.
- Visualize distributions using histograms, bar plots, and line charts.
- Compare crime rates across states and years.

Step 3: Correlation and Pattern Detection

- Generate correlation matrices for offense types and property loss.
- Examine relationships between drugs and violent crimes.
- Explore demographic influences on crime rates.

Step 4: Temporal and Spatial Analysis

- Time-series plots to detect trends over 2000–2022.
- State-level comparisons to identify high-risk regions.
- Seasonal or monthly patterns analysis (if relevant).

Step 5: Insights and Interpretation

- Summarize trends in crime prevalence and offender/victim demographics.

- Highlight correlations between crime types, property loss, and other variables.
- Discuss potential social, policy, or environmental factors influencing patterns.

10. Expected Outcomes

- Cleaned, structured, and documented the dataset for analysis.
- Visualizations of trends, correlations, and state-level comparisons.
- Insights into demographic and behavioral patterns in U.S. crime.
- Identification of high-risk states, crime categories, and years.
- Clear, data-driven recommendations or discussion points for policy or research.

11. Significance of the Study

This analysis provides valuable insights for:

- Law enforcement agencies: Resource allocation and targeted interventions.
- Policymakers: Evidence-based policy decisions.
- Researchers and social scientists: Understanding social and demographic influences on crime.